Lab 6

Cities and Transit

Network Distances, Isochrone Maps

DATA 20195 Spring 2026

Emma Ross and Michael Xiao

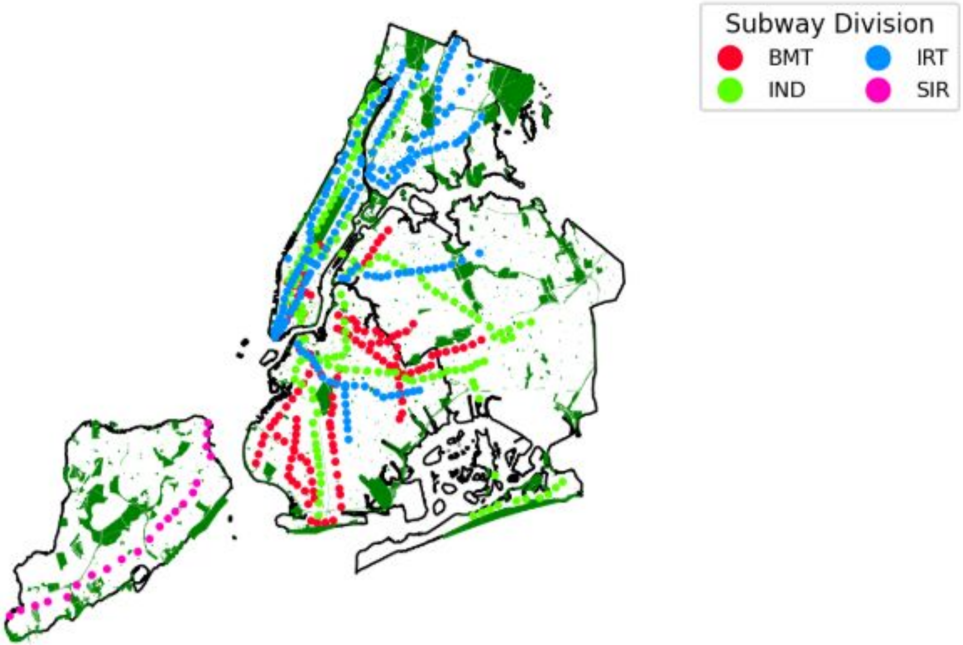
Now you're in New York!

Today we will explore data from the **NYC Open Data Portal** on *subway stations* and *parks*.

Our notebook today is computationally slow and intensive, so let's jump right in.

Images credit: Wikipedia

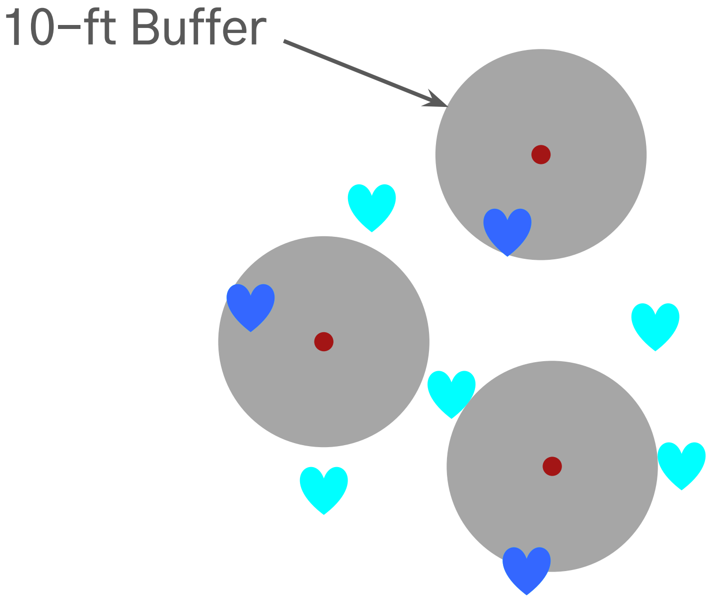
Data Visualization Clinic



What are some improvements you made?

Buffering

`red_points_gdf.buffer(10)`



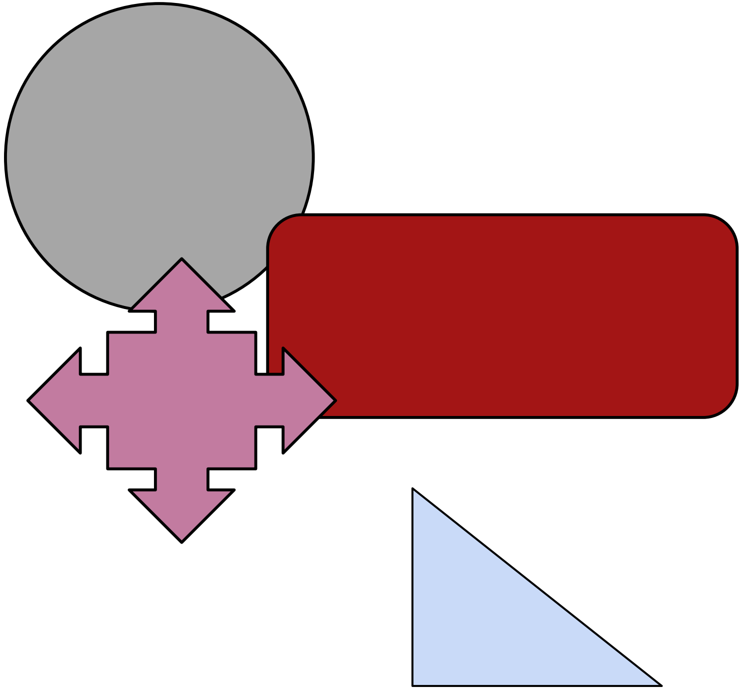
Which s are within 10 feet of a point?

Dissolving

`shapes_gdf.dissolve()`

ID	Name	Geometry
1	circle	Polygon(...)
2	multi_pointer	Polygon(...)
3	rounded_rectangle	Polygon(...)
4	triangle	Polygon(...)

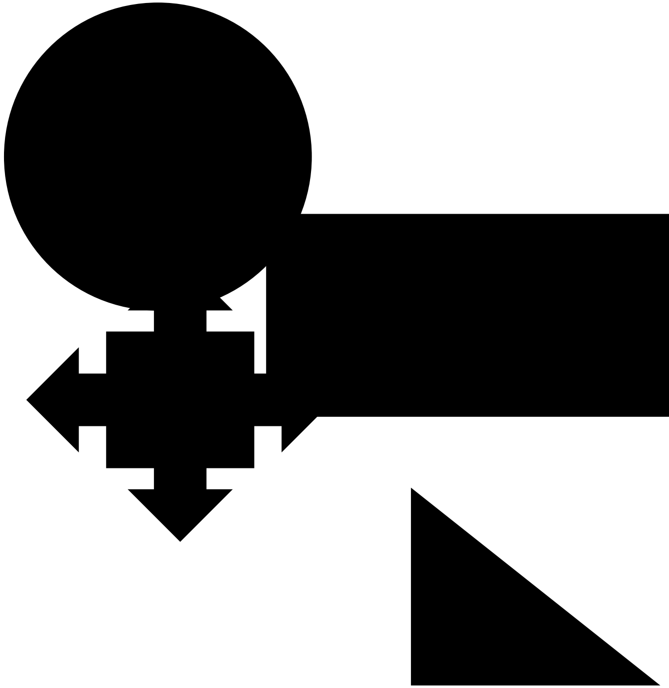
`shapes_gdf`



Dissolving

`shapes_gdf.dissolve()`

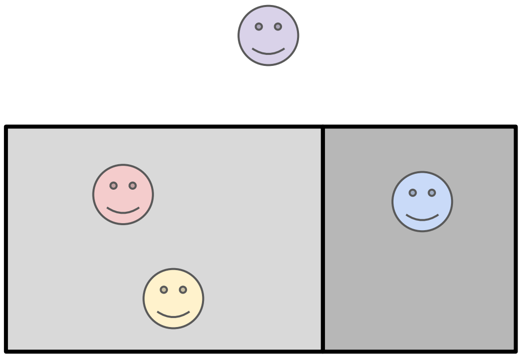
ID	Geometry
1	MultiPolygon(...)



Spatial Join `(sjoin)`

SID	Year	Major	GPA	Geometry
1	Senior	Data Science	3.7	Point(...)
2	Junior	Political Science	3.8	Point(...)
3	Senior	Mathematics	4.0	Point(...)
4	Sophomore	Data Science	3.7	Point(...)

`student_gdf`

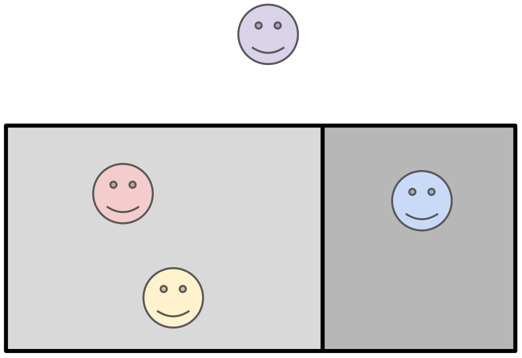


Room ID	Course	Credits	Geometry
A	Data and the State	300	Polygon(...)
B	Seminar	150	Polygon(...)

`room_gdf`

```
student_gdf.sjoin(room_gdf, how="left")
```

Spatial Join (`sjoin`)



```
student_gdf.sjoin(room_gdf, how="left")
```

SID	Year	Major	GPA	Geometry	Room ID	Course	Credits
1	Senior	Data Science	3.7	Point(...)	A	Data and the State	300
2	Junior	Political Science	3.8	Point(...)	A	Data and the State	300
3	Senior	Mathematics	4.0	Point(...)	B	Seminar	150
4	Sophomore	Data Science	3.7	Point(...)	None	None	None

Discussion 1: Straight Lines

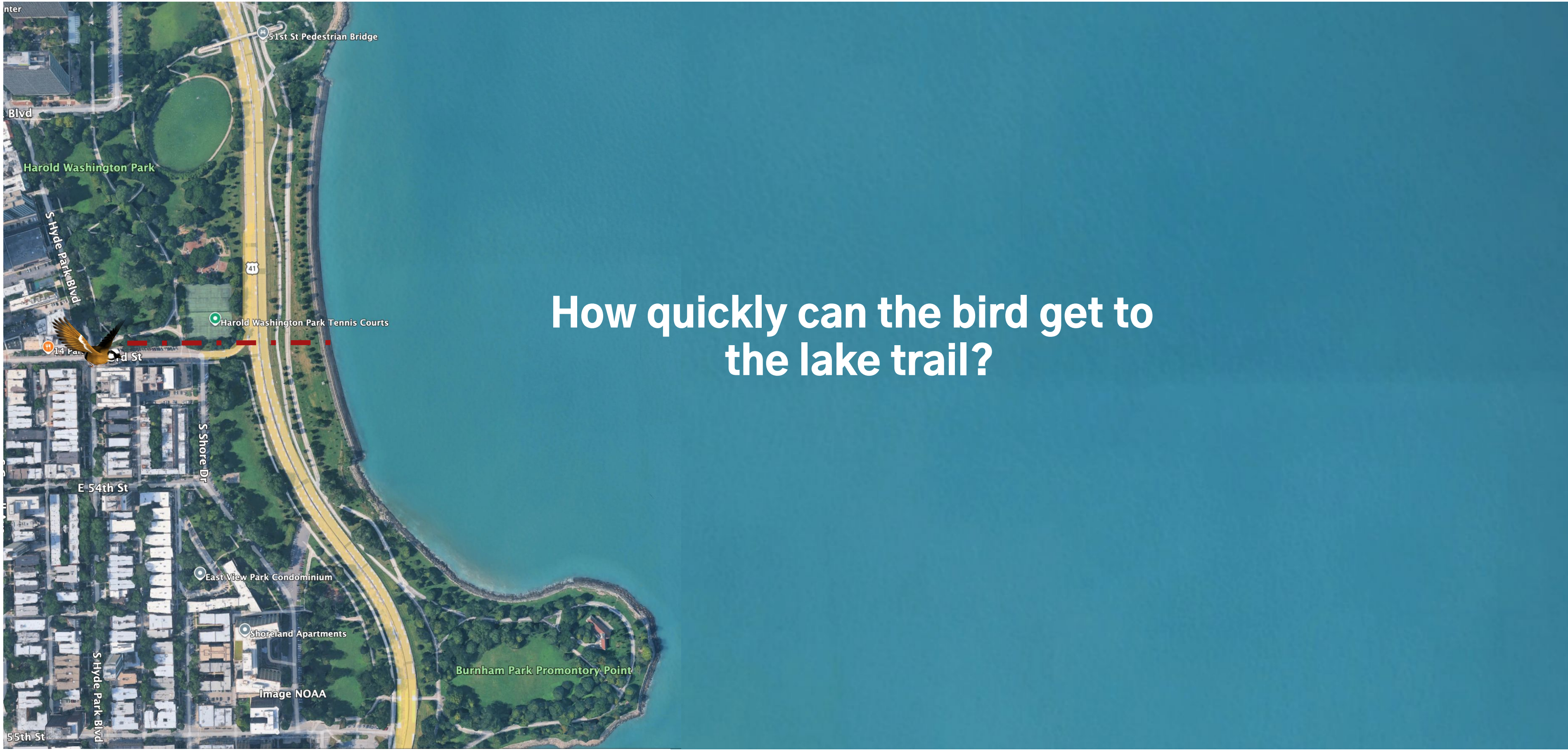


Image credit: Google Earth

Discussion 1: Straight Lines

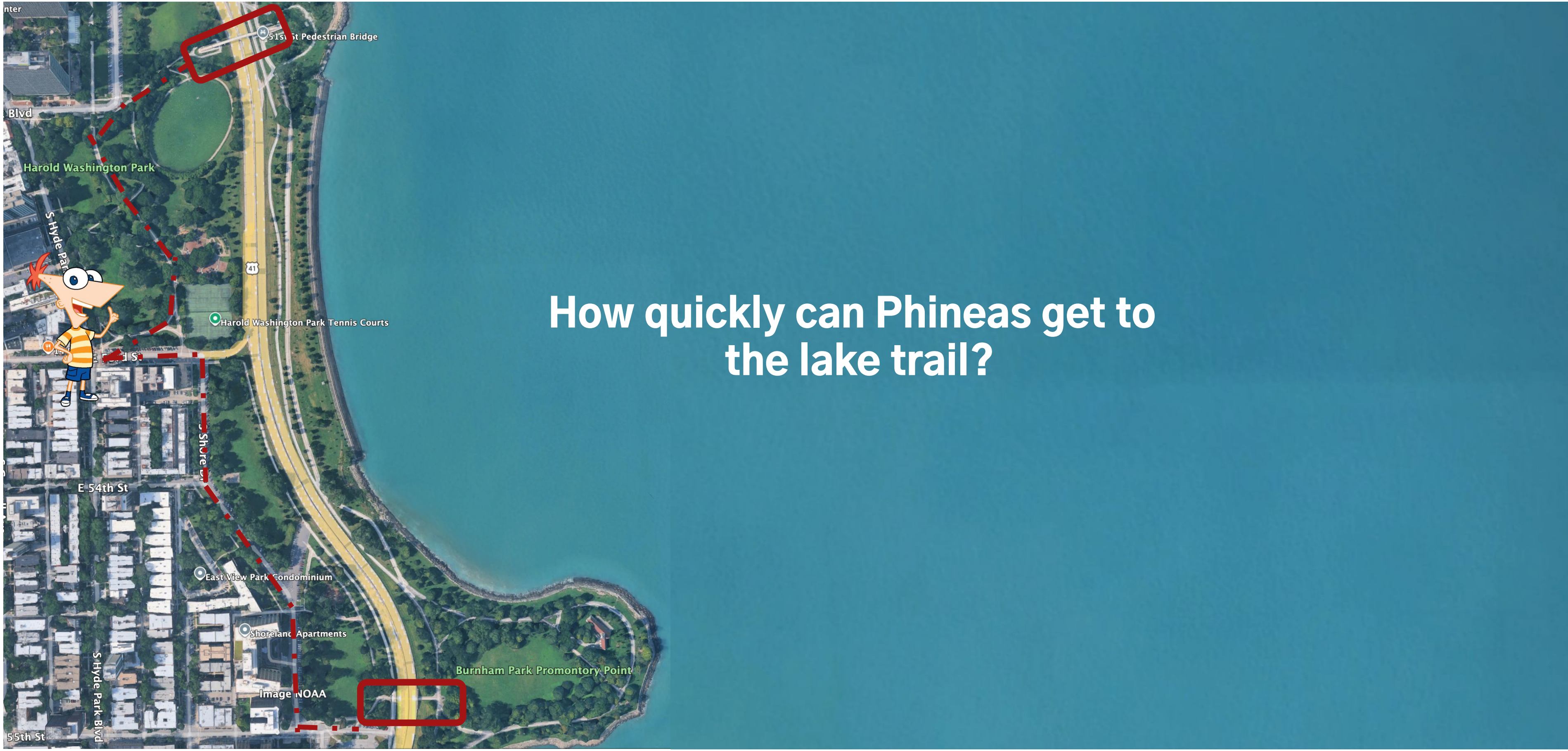


Image credit: Google Earth

Discussion 1: Straight Lines

Creating a buffer assumes straight–line distance is sufficient.

1. **What assumptions does it make about how people actually move through a city?**

In Task 2 we find that Brooklyn has the most subway stations farther than a 1/4 mile (straight–line) from a park.

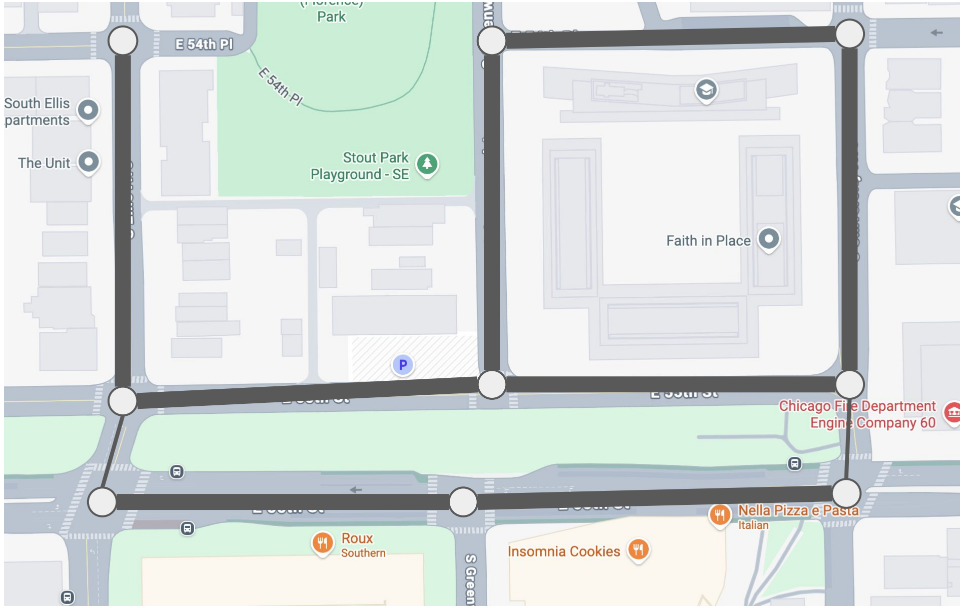
2. **Does this mean Brooklyn is the borough with the greatest need for more parks?**

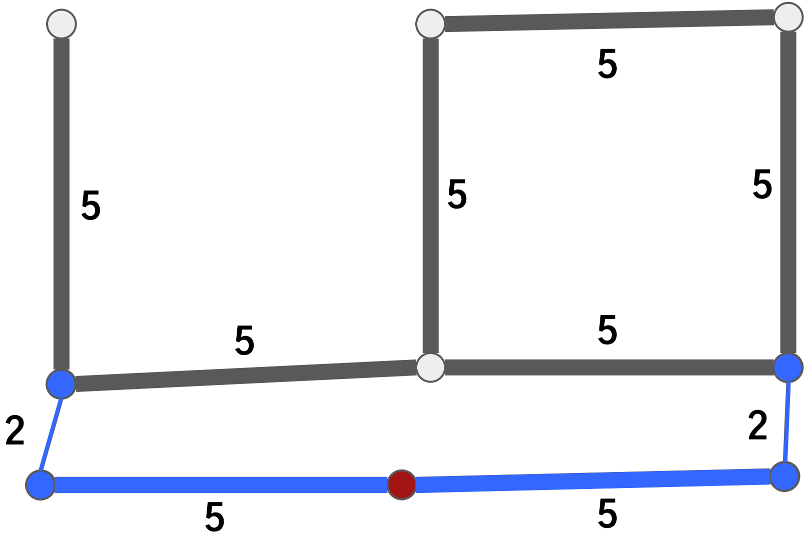
What other denominators might matter: population, land area, total park acreage, or something else.

Parks by borough

borough	
B	628
Q	476
X	398
M	395
R	161

Street Network



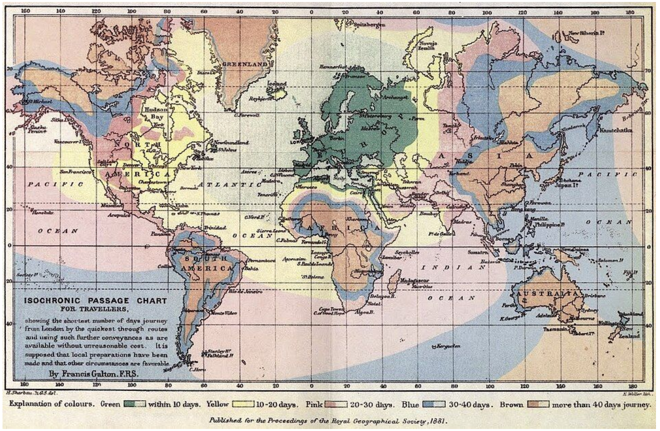


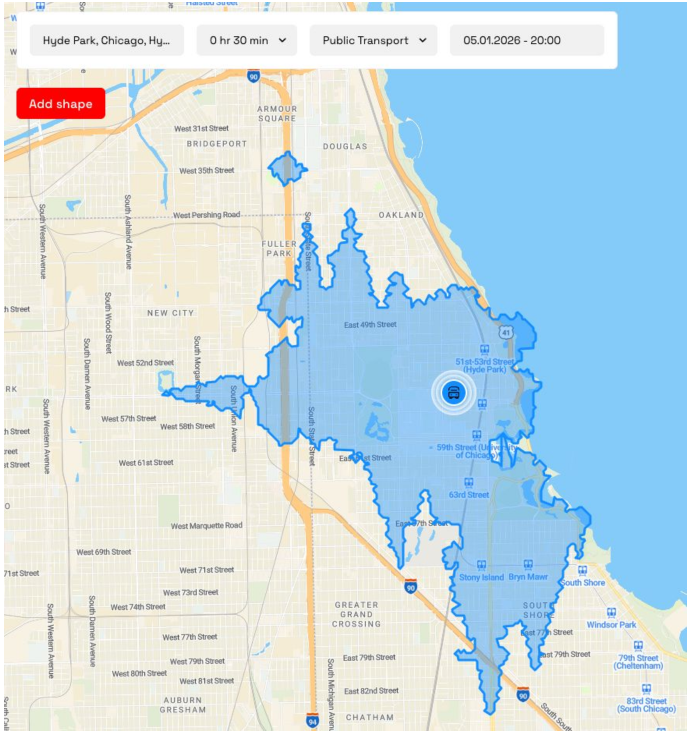
Starting **here**, where can I reach within a 10 minute walk?

Image credit: Google Maps

Isochrone Maps

- iso = equal   khronos = time
- map that shows, either with **shading**, or by **highlighting individual paths** the areas or points of interest that are **reachable within a certain time**
- Francis Galton made the first isochrone map in **1881** to show areas **reachable from London** within certain time limits





https://app.traveltimeplatform.com/

Image credit: Wikimedia https://commons.wikimedia.org/wiki/File:Isochronic_Passage_Chart_Francis_Galton_1881.jpg

Discussion 2:

What are some differences you notice in the **spread** and **distribution** of the buffers based on **transportation type** and **time allowance**?

If you were a **team of city planners**, how would you consider all of this information in decisions on new roads, new parks, bike lanes, etc.?